

CUSTOMER CHURN ANALYSIS

1. Project Overview

- The purpose of this analysis is to understand customer churn by examining customer demographics, tenure, engagement behavior, support interactions, payment behavior, subscription type, contract length, and customer value.
- The analysis aims to identify customer characteristics and behavioral patterns associated with higher churn rates, recognize high-risk customer segments, and evaluate the potential business impact of customer churn.
- The findings can help the business prioritize customer retention efforts and support more data-driven decision-making.

2. Dataset Summary

- **Rows:** 440,832 customer records after removing one completely empty row
- **Columns:** 12
- **Key Features:** The dataset includes information related to:
 - **Customer Information**
 - Customer ID, Age, Gender
 - **Customer Relationship**
 - Tenure, Usage Frequency, Last Interaction
 - **Customer Support & Payment Behavior**
 - Support Calls, Payment Delay
 - **Subscription Information**
 - Subscription Type, Contract Length
 - **Customer Value**
 - Total Spend
 - **Churn Information**
 - Churn status
- **Missing Data:**

One completely empty row was identified and removed. No remaining missing values were detected after data cleaning.
- **Duplicate Data:**

No duplicate records were detected.
- **Churn Distribution:**

- Churned Customers: 249,999
- Retained Customers: 190,833
- Churn Rate: 56.71%
- Retention Rate: 43.29%

The dataset therefore shows a relatively high overall churn rate, making customer retention a key business concern.

3. Exploratory Data Analysis using Python

We began with data preparation, data quality validation, and exploratory analysis in Python.

- Data Loading:

Imported the raw customer churn dataset using **pandas**.

- Initial Exploration:

Used `df.info()` to examine the dataset structure, data types, and non-null values, and used `df.describe()` to review the distribution and summary statistics of numerical variables.

```
<class 'pandas.DataFrame'>
RangeIndex: 440832 entries, 0 to 440832
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            440832 non-null float64
1   Age                   440832 non-null float64
2   Gender                440832 non-null str
3   Tenure                440832 non-null float64
4   Usage Frequency       440832 non-null float64
5   Support Calls         440832 non-null float64
6   Payment Delay         440832 non-null float64
7   Subscription Type     440832 non-null str
8   Contract Length       440832 non-null str
9   Total Spend           440832 non-null float64
10  Last Interaction       440832 non-null float64
11  Churn                 440832 non-null float64
dtypes: float64(9), str(3)
memory usage: 40.4 MB
```

	CustomerID	Age	Tenure	Usage Frequency	Support Calls	Payment Delay	Total Spend	Last Interaction	Churn
count	440832.000000	440832.000000	440832.000000	440832.000000	440832.000000	440832.000000	440832.000000	440832.000000	440832.000000
mean	225398.667955	39.373153	31.256336	15.807494	3.604437	12.965722	631.616223	14.480868	0.567107
std	129531.918550	12.442369	17.255727	8.586242	3.070218	8.258063	240.803001	8.596208	0.495477
min	2.000000	18.000000	1.000000	1.000000	0.000000	0.000000	100.000000	1.000000	0.000000
25%	113621.750000	29.000000	16.000000	9.000000	1.000000	6.000000	480.000000	7.000000	0.000000
50%	226125.500000	39.000000	32.000000	16.000000	3.000000	12.000000	661.000000	14.000000	1.000000
75%	337739.250000	48.000000	46.000000	23.000000	6.000000	19.000000	830.000000	22.000000	1.000000
max	449999.000000	65.000000	60.000000	30.000000	10.000000	30.000000	1000.000000	30.000000	1.000000

- Missing Value Analysis:

Checked missing values across all columns and identified one completely empty row. The empty row was removed before further analysis to ensure accurate customer-level calculations.

- Duplicate Check:

Checked for duplicate records and duplicate customer IDs.

No duplicate customer records were identified.

- **Data Type Validation:** Validated and converted numerical columns such as:

- Age
- Tenure
- Usage
- Frequency
- Support Calls
- Payment Delay
- Total Spend
- Last Interaction
- Churn

Categorical fields were also standardized by removing unnecessary whitespace.

- **Column Standardization:**

Renamed columns to **snake case** for better readability, consistency, and database compatibility.

For example:

CustomerID	→ customerid
Usage Frequency	→ usage_frequency
Support Calls	→ support_calls
Payment Delay	→ payment_delay
Total Spend	→ total_spend

- **Data Quality Validation:**

Validated business ranges for numerical variables and checked for invalid or negative values.

After cleaning, the dataset was exported as a cleaned CSV file.

- **Database Integration:**

Connected Python to PostgreSQL using SQLAlchemy and loaded the cleaned dataset into the customer_churn table for SQL-based analysis.

4. Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer key business questions related to customer churn.

4.1. Overall Customer Churn Rate

Measure the overall churn and retention performance of the customer base.

The analysis identified:

total_customers	churned_custom...	retained_custom...	churn_rate	retention_rate
abc Filter...	abc Filter...	abc Filter...	abc Filter...	abc Filter...
440832	249999	190833	56.71	43.29

→ The dataset contains **440,832 customers**, of which **249,999 customers churned**. This results in an overall churn rate of **56.71%**, compared with a retention rate of **43.29%**.

⇒ The high churn rate indicates that customer retention is an important area for the business to monitor and improve.

4.2. Customer Demographics and Churn

Analyze churn differences across customer demographic groups.

Churn Rate by Gender

gender	total_customers	churned_custom...	churn_rate
abc Filter...	abc Filter...	abc Filter...	abc Filter...
Female	190580	127058	66.67
Male	250252	122941	49.13

→ Female customers recorded a higher churn rate of **66.67%**, compared with **49.13%** for male customers.

Churn Rate by Age Group

age_group	total_customers	churned_custom...	churn_rate
abc Filter...	abc Filter...	abc Filter...	abc Filter...
56-65	54425	54425	100.00
46-55	90203	52178	57.85
18-25	73683	41576	56.43
26-35	104108	51139	49.12
36-45	118413	50681	42.80

→ The **56–65** age group had the highest observed churn rate at **100.00%**, while the **36–45** group had the lowest rate at **42.80%**.

⇒ These results show that churn rates vary across demographic groups. However, these findings represent associations in the dataset and should not be interpreted as evidence that demographic characteristics directly cause churn.

4.3. Customer Tenure and Churn

This analysis evaluates whether churn rates vary according to how long customers have been with the company.

Churn Rate by Tenure Group

tenure_group	total_customers	churned_custom...	churn_rate
abc Filter...	abc Filter...	abc Filter...	abc Filter...
13-24 months	79967	50597	63.27
1-12 months	85129	50000	58.73
49-60 months	92331	50133	54.30
37-48 months	91932	49760	54.13
25-36 months	91473	49509	54.12

→ The **13–24 month** tenure group recorded the highest churn rate at **63.27%**.

The average tenure was also slightly lower among churned customers:

customer_status	total_customers	avg_tenure
abc Filter...	abc Filter...	abc Filter...
Churned	249999	30.47
Retained	190833	32.28

⇒ The results indicate that churn is not limited to newly acquired customers, with the **13–24 month** group showing the highest churn rate among the tenure segments.

4.4. Customer Engagement and Churn

This analysis examines usage frequency and the number of days since the customer's last interaction.

usage_frequenc...	total_customers	churned_custom...	churn_rate
abc Filter...	abc Filter...	abc Filter...	abc Filter...
1-5	68538	42151	61.50
6-10	69886	41787	59.79
26-30	75783	41677	55.00
21-25	75557	41546	54.99
11-15	75728	41612	54.95
16-20	75340	41226	54.72

→ Customers with **1–5 usage events** had the highest churn rate at **61.50%**.

The average usage frequency was:

customer_status	total_customers	avg_usage_freq...
abc Filter...	abc Filter...	abc Filter...
Churned	249999	15.46
Retained	190833	16.26

Churn Rate by Last Interaction

last_interaction_...	total_customers	churned_custom...	churn_rate
abc Filter...	abc Filter...	abc Filter...	abc Filter...
21-25 days	63475	42313	66.66
26-30 days	63585	42278	66.49
16-20 days	63377	42045	66.34
11-15 days	83375	41160	49.37
6-10 days	83639	41179	49.23
1-5 days	83381	41024	49.20

→ Customers with **16–30 days since their last interaction** recorded churn rates of approximately **66%**, compared with approximately **49%** among customers with 1–15 days since their last interaction.

⇒ This indicates that lower usage and longer periods since the last interaction are associated with higher observed churn rates.

4.5. Customer Support Activity and Churn

This analysis evaluates whether the number of customer support calls is associated with churn.

Average Support Calls

customer_status	total_customers	avg_support_calls
abc Filter...	abc Filter...	abc Filter...
Churned	249999	5.14
Retained	190833	1.59

→ Churned customers had an average of **5.14 support calls**, compared with **1.59** among retained customers.

Churn Rate by Support Calls

support_calls_gr...	total_customers	churned_custom...	churn_rate
abc Filter...	abc Filter...	abc Filter...	abc Filter...
7-8	47429	47429	100.00
9-10	47530	47530	100.00
5-6	48472	47155	97.28
3-4	91479	44626	48.78
0-2	205922	63259	30.72

→ The churn rate increased substantially among customers with more support interactions. Customers with **7–10 support calls** had an observed churn rate of **100.00%**.

⇒ This makes frequent support interaction an important indicator to monitor when identifying potentially high-risk customers.

4.6. Payment Behavior and Churn

This analysis examines whether payment delays are associated with customer churn.

Average Payment Delay

customer_status	total_customers	avg_payment_d...
abc Filter...	abc Filter...	abc Filter...
Churned	249999	15.22
Retained	190833	10.02

→ Churned customers had an average payment delay of **15.22 days**, compared with **10.02 days** among retained customers.

Churn Rate by Payment Delay

payment_delay_...	total_customers	churned_custom...	churn_rate
abc Filter...	abc Filter...	abc Filter...	abc Filter...
21-25 days	42134	42134	100.00
26-30 days	41896	41896	100.00
6-10 days	84793	39502	46.59
11-15 days	85571	39860	46.58
16-20 days	84984	39484	46.46
0-5 days	101454	47123	46.45

→ Customers with payment delays of **21–30 days** had an observed churn rate of **100.00%**.

⇒ The results suggest that significant payment delays are strongly associated with higher observed churn in this dataset.

4.7. Subscription and Contract Type Analysis

This analysis compares churn rates across subscription types and contract lengths.

Churn Rate by Subscription Type

subscription_type	total_customers	churned_custom...	churn_rate
abc Filter...	abc Filter...	abc Filter...	abc Filter...
Basic	143026	83210	58.18
Standard	149128	83616	56.07
Premium	148678	83173	55.94

→ The **Basic** subscription type had the highest churn rate at **58.18%**, although the difference between subscription types was relatively small.

Churn Rate by Contract Length

contract_length	total_customers	churned_custom...	churn_rate
 Filter...	 Filter...	 Filter...	 Filter...
Monthly	87104	87104	100.00
Annual	177198	81646	46.08
Quarterly	176530	81249	46.03

→ Contract length showed a much larger difference than subscription type.

⇒ Customers with a **Monthly** contract had an observed churn rate of **100.00%**, compared with approximately **46%** for both Annual and Quarterly contracts.

4.8. High-Risk Customer Segments

This analysis combines **Contract Length**, **Support Risk**, **Payment Risk**, and **Engagement Risk** to identify customer segments with high churn rates.

Customer Segments with High Churn Rates

contract_length	support_risk	payment_risk	engagement_risk	total_customers	churned_custom...	churn_rate
 Filter...	 Filter...	 Filter...	 Filter...	 Filter...	 Filter...	 Filter...
Monthly	Low Support Calls	Low Payment Delay	Normal Usage	12897	12897	100.00
Monthly	Low Support Calls	High Payment Delay	Low Usage	6836	6836	100.00
Monthly	High Support Calls	Low Payment Delay	Low Usage	7578	7578	100.00
Monthly	Low Support Calls	High Payment Delay	Normal Usage	13709	13709	100.00
Monthly	High Support Calls	High Payment Delay	Normal Usage	16177	16177	100.00
Monthly	High Support Calls	High Payment Delay	Low Usage	8161	8161	100.00
Monthly	Low Support Calls	Low Payment Delay	Low Usage	6434	6434	100.00
Monthly	High Support Calls	Low Payment Delay	Normal Usage	15312	15312	100.00
Quarterly	High Support Calls	High Payment Delay	Low Usage	8219	8183	99.56
Annual	High Support Calls	High Payment Delay	Low Usage	8257	8216	99.50
Quarterly	High Support Calls	High Payment Delay	Normal Usage	16394	16273	99.26
Annual	High Support Calls	High Payment Delay	Normal Usage	16546	16391	99.06
Quarterly	High Support Calls	Low Payment Delay	Low Usage	7759	7652	98.62
Annual	High Support Calls	Low Payment Delay	Low Usage	7720	7606	98.52
Quarterly	High Support Calls	Low Payment Delay	Normal Usage	15549	15188	97.68
Annual	High Support Calls	Low Payment Delay	Normal Usage	15759	15377	97.58
Quarterly	Low Support Calls	High Payment Delay	Low Usage	14077	6347	45.09
Annual	Low Support Calls	High Payment Delay	Low Usage	14207	6350	44.70
Annual	Low Support Calls	High Payment Delay	Normal Usage	31689	12438	39.25
Quarterly	Low Support Calls	High Payment Delay	Normal Usage	31757	12421	39.11
Annual	Low Support Calls	Low Payment Delay	Low Usage	24661	5325	21.59
Quarterly	Low Support Calls	Low Payment Delay	Low Usage	24515	5250	21.42
Quarterly	Low Support Calls	Low Payment Delay	Normal Usage	58260	9935	17.05
Annual	Low Support Calls	Low Payment Delay	Normal Usage	58359	9943	17.04

The results show that customers with **Monthly contracts** represent the highest-risk segments, with several combinations recording a **100.00% churn rate**.

Among customers with **Quarterly or Annual contracts**, those with **High Support Risk, High Payment Risk, and Low Engagement Risk** also recorded extremely high churn rates of **99.56%** and **99.50%**, respectively.

The customer-level output contains **249,999 churned customers**, with each customer assigned a **Risk Score from 0 to 5**, providing a basis for identifying customers with different levels of churn risk.

4.9. Business Impact Analysis

This analysis evaluates the financial loss resulting from customer churn and examines the prevalence of key risk factors across the total customer base.

Financial Loss from Churn

churned_custom...	total_spend_lost	avg_spend_per...	
abc Filter...	abc Filter...	abc Filter...	
249999	135320840.60	541.29	

→ A total of 249,999 customers churned, resulting in an overall revenue loss of \$135,320,840.60.

⇒ The average spend lost per churned customer was \$541.29.

Prevalence of Key Risk Factors

total_customers	monthly_contra...	high_support_ca...	high_payment_d...
abc Filter...	abc Filter...	abc Filter...	abc Filter...
440832	19.76	21.54	19.06

→ Across the 440,832 total customers, high-risk operational and contract attributes each account for roughly one-fifth of the user base.

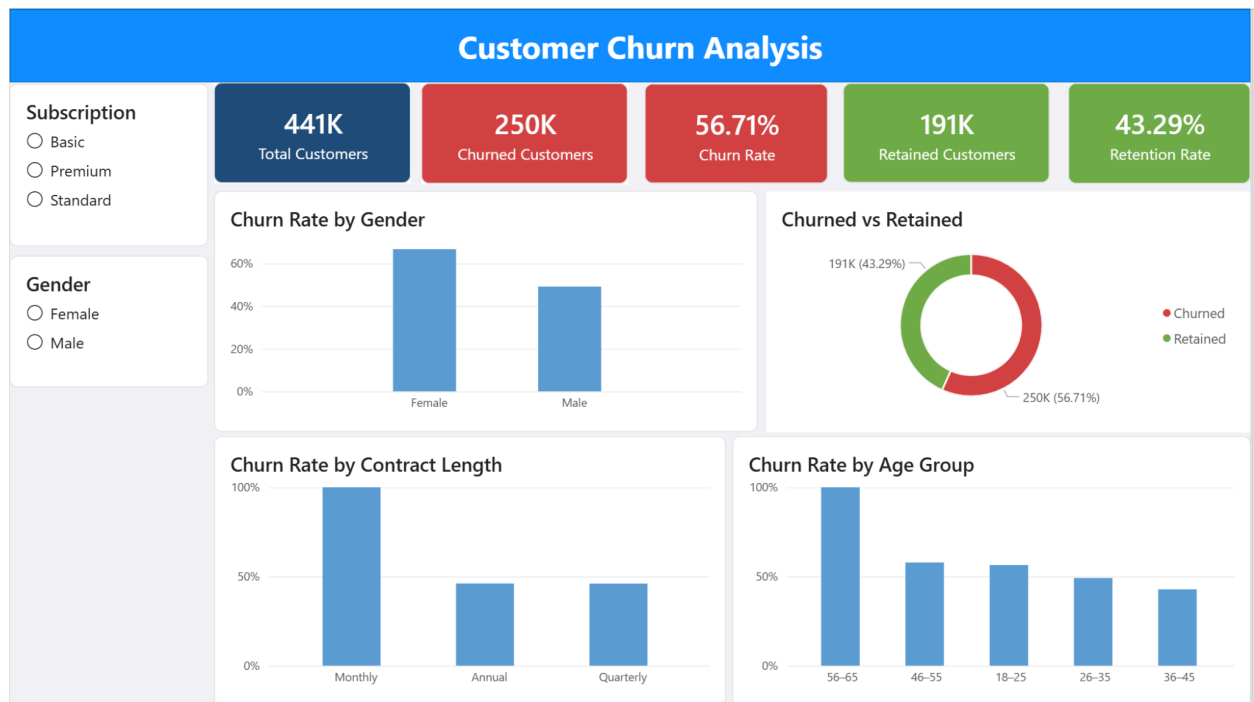
⇒ High support calls represented the largest risk segment at 21.54%, followed by monthly contracts at 19.76% and high payment delays at 19.06%.

5. Dashboard in Power BI

Finally, we built an interactive **Power BI dashboard** to present the analysis visually across three analytical pages.

Page 1 — Executive Overview

The first page provides a high-level overview of customer churn performance.



Key components include:

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate
- Retention Rate
- Churned vs Retained
- Churn Rate by Gender

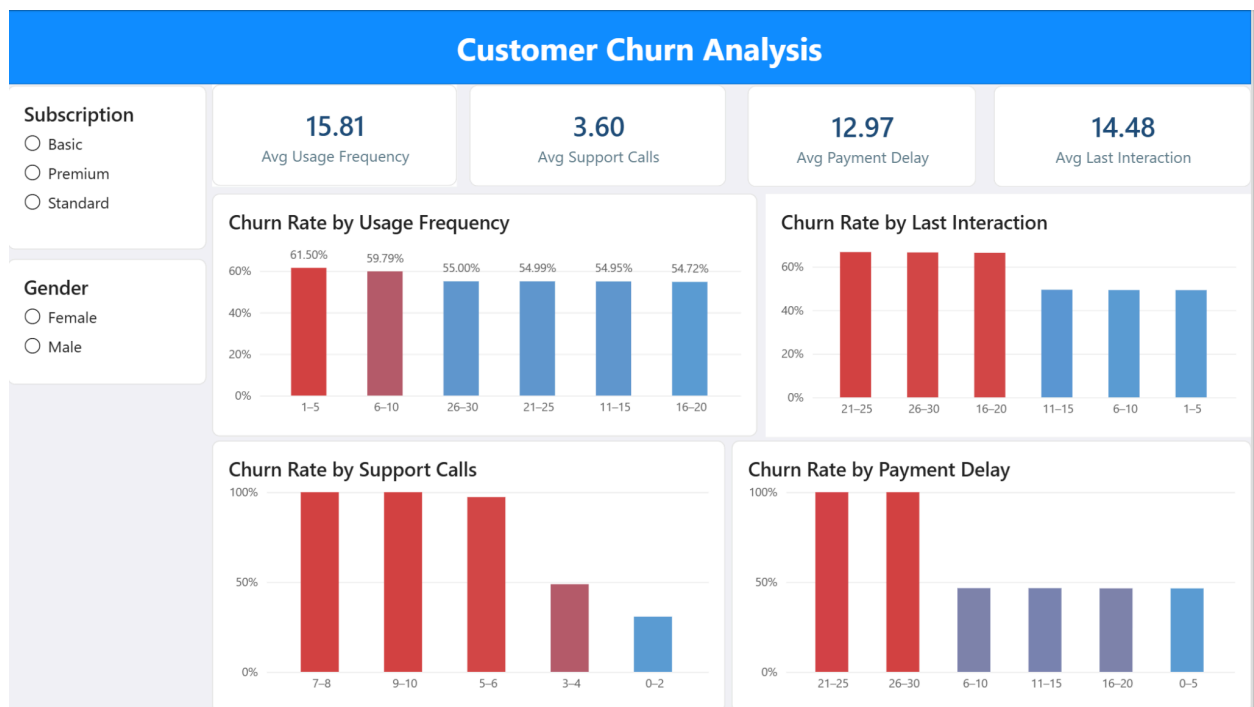
- Churn Rate by Age Group
- Churn Rate by Contract Length

This page answers:

What is happening with customer churn?

Page 2 — Customer Behavior & Churn Drivers

The second page focuses on behavioral factors associated with churn.



Key components include:

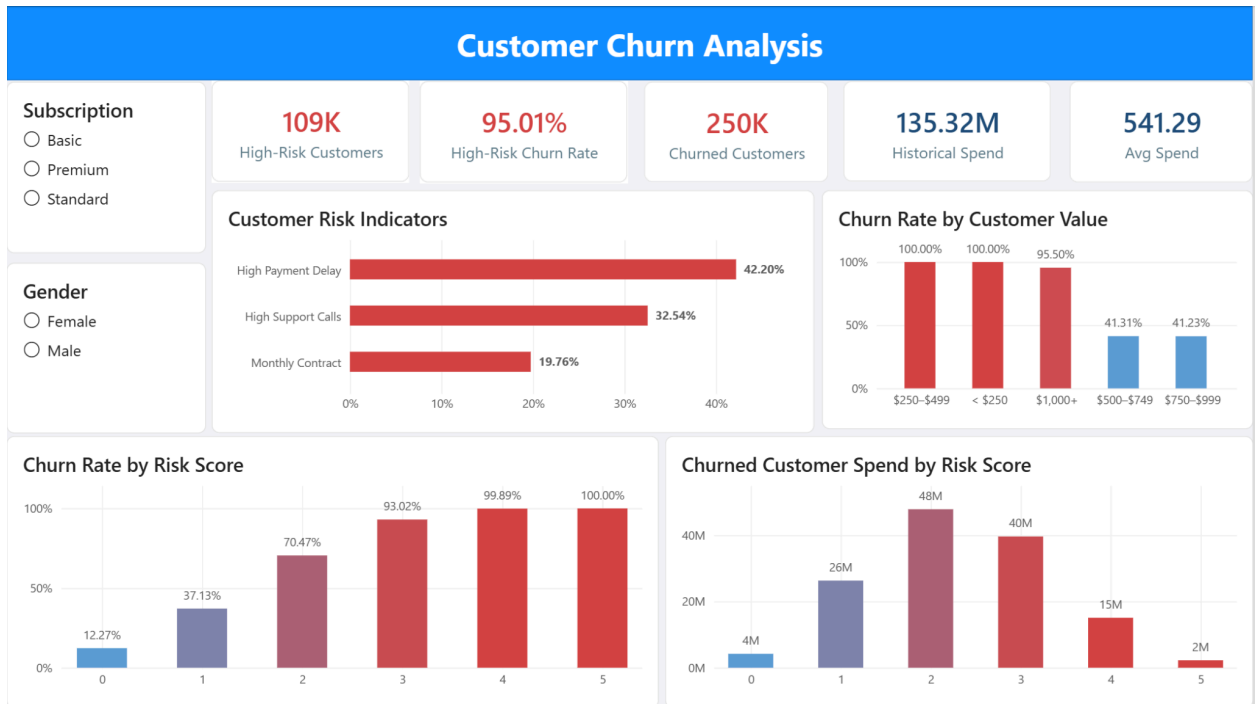
- Average Usage Frequency
- Average Support Calls
- Average Payment Delay
- Average Last Interaction
- Churn Rate by Usage Frequency
- Churn Rate by Last Interaction
- Churn Rate by Support Calls
- Churn Rate by Payment Delay

This page answers:

What customer behaviors are associated with higher churn?

Page 3 — High-Risk Customers & Business Impact

The third page focuses on identifying high-risk customers and evaluating the potential business impact.



Key components include:

- High-Risk Customers
- High-Risk Churn Rate
- Churned Customers
- Historical Spend
- Average Spend of Churned Customers
- Customer Risk Indicators
- Churn Rate by Customer Value
- Churn Rate by Risk Score
- Churned Customer Spend by Risk Score

This page answers:

Who is most at risk, and what is the potential business impact?

⇒ Together, the three dashboard pages provide a progression from **overall performance** → **churn drivers** → **risk prioritization and business impact**.

6. Business Recommendations

Based on the analysis results, the following recommendations are proposed:

- **Prioritize high-risk customers** by using the Risk Score to identify customers with multiple churn-related indicators.
- **Focus on customers with frequent support interactions**, particularly customers with five or more support calls, and investigate the underlying issues causing repeated support contacts.
- **Monitor payment delays** and consider proactive communication or payment-support programs for customers showing significant payment delays.
- **Improve engagement among low-usage customers** by using targeted communication, personalized offers, or product education to encourage greater service usage.
- **Pay particular attention to monthly-contract customers**, as this segment shows a substantially higher churn rate than quarterly and annual contract customers.
- **Strengthen retention efforts for high-value customers**, especially when high customer value is combined with multiple churn-risk indicators.
- **Use the Power BI dashboard for ongoing monitoring** of churn rates, behavioral indicators, risk segments, and customer value to support data-driven retention decisions.
- **Further investigate high-risk segments** before implementing retention campaigns, since the analysis identifies associations with churn rather than proving that these factors directly cause customers to leave.

Overall Business Conclusion

The analysis shows that customer churn is a significant business challenge, with an overall churn rate of **56.71%**.

Several factors are strongly associated with higher churn in this dataset, particularly:

- **Monthly contracts**

- **High support-call frequency**
- **Long payment delays**
- **Low usage frequency**
- **Longer periods since the last interaction**
- **Higher combined Risk Scores**

Most importantly, the Risk Score analysis shows a clear increase in churn as the number of risk indicators increases, from **12.27% churn at Risk Score 0 to 100% at Risk Score 5**.

Therefore, the business can use a combination of **customer behavior, service interaction, payment behavior, contract information, and customer value** to prioritize retention efforts and focus resources on customers with the highest observed risk.